5 Fig. 5.1 shows some equipotential lines around an electricity transmission cable at +200 kV.

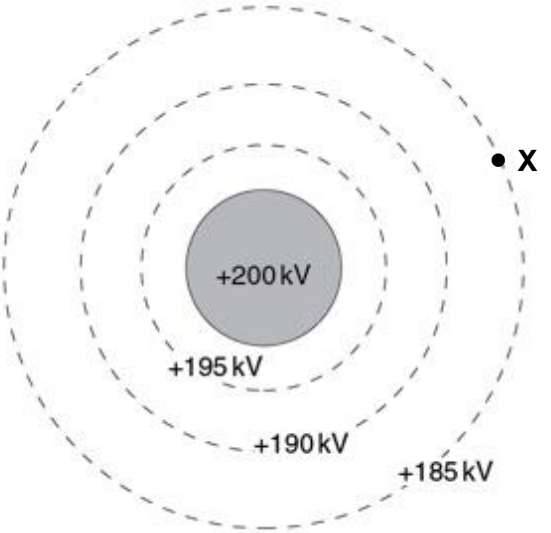

Fig. 5.1

(a) State the feature of the diagram which shows that the electric field strength decreases with distance from the transmission cable.

.....

.....

.....

..... [2]

(b) The 195 kV equipotential line is 5.0 mm from the surface of the 200 kV transmission cable. Use this information to estimate the electric field strength at the surface of the cable.

electric field strength = ... $V\ m^{-1}$ [2]

(c) An electron is released from rest at point X. Determine the speed of the electron when it reaches the surface of the transmission cable.

speed = ... $m\ s^{-1}$ [2]

(d) Fig. 5.2 shows the transmission cable surrounded on three sides by an earthed metal shield at zero potential.

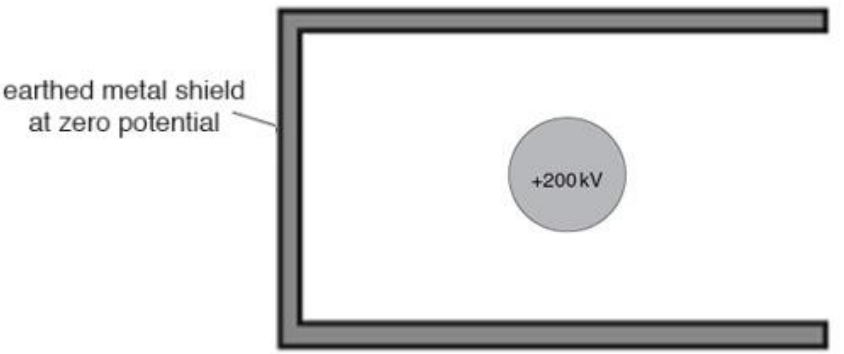

Fig. 5.2

On Fig. 5.2, sketch the shape of the electric field within and beyond the shield by drawing field lines from the cable to the shield and in the space beyond the open end.

[3]